

emerging sectors that later became globally dominant. Since the 1990s, global manufacturing shifted rapidly towards electronics, semiconductors, industrial machinery, robotics, synthetic materials, batteries and advanced components. China, South Korea and Taiwan invested aggressively in these sectors early. India largely missed that transition.

Instead, India's economic success increasingly came from services. These sectors benefited from lower Indian wages compared to the US, earned large amounts of foreign exchange and helped create a growing middle class. But they also reduced the pressure to build strong manufacturing industries, leading many experts to describe India as a services-led economy.

98 | Many Indian firms discovered that importing products from China and selling them domestically was easier and more profitable than manufacturing them locally. Gradually, trading replaced manufacturing across many sectors. Even in pharmaceuticals, many Indian companies reduced domestic production of bulk drugs and APIs and increasingly focused on making higher-value formulations using cheaper Chinese APIs, increasing dependence on imports from China.

Third, India became an expensive place to manufacture.

Industrial power tariffs remained high. Logistics costs stayed elevated because of fragmented supply chains and expensive transport systems. Lending rates for manufacturers remained 9–10%, compared with 4–5% in China. Compliance systems became extremely complex, especially for MSMEs. Environmental clearances, tax disputes, inspections, land approvals and changing regulations created uncertainty and delays.

India also developed distorted tariff structures. In several industries, import duties on raw materials and components became higher than duties on finished products. This inverted duty structure made domestic manufacturing commercially irrational. It became cheaper to import finished goods than produce them locally.

The result was shallow industrialisation.

India expanded assembly operations, particularly in smartphones, solar equipment and electronics. But

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much of the value addition remained outside India because critical components continued to be imported.

Meanwhile, traditional labour-intensive sectors where India once had advantages—textiles, garments, light manufacturing—steadily lost competitiveness to Vietnam and Bangladesh.

China moved in the opposite direction.

While India debated reforms, China built industrial ecosystems at massive scale. China did not simply build high scale and hi technology factories. It built ports, logistics systems, industrial clusters, supplier networks and technical training institutions.

Seven of the world's ten busiest ports are Chinese. Customs systems became digital and fast. Bureaucrats were rewarded for enabling manufacturing growth rather than blocking it.

China also thought strategically about future supply chains. In the 1990s and 2000s, it acquired access to lithium, cobalt, and rare-earth resources across the world—long before EVs and clean energy became major industries. Those investments now underpin China's dominance in batteries, solar panels and advanced electronics.

India still has time to correct course, but the policy approach must fundamentally change.

The first priority is to move towards real ecosystem building. A semiconductor plant, a textile cluster and a machinery industry each require different support systems—land, logistics, skilled workers, suppliers and financing. Industrial policy must become sector-specific rather than generic.

Second, India must sharply reduce the cost of manufacturing. Competitive electricity tariffs, lower logistics costs, affordable industrial finance and stable regulations matter more than subsidies.

Third, India must recognise the importance of downstream manufacturing. Policy should no longer disproportionately favour upstream commodity sectors at the expense of MSME value addition industries.

Fourth, India should launch a large-scale reverse engineering and technology localisation programme for imported industrial products such as motors, pumps, bearings, machine tools and industrial equipment.

Finally, India's challenge today is the absence of coordinated execution that builds real industrial depth. A technically competent leader for such initiatives can deliver on this front. **BT**

Views are personal